

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

5. A company of animal feeds produces k different types of feed by mixing r different types of raw materials (seeds and flours). In order to obtain a unit of feed i the factory needs a_{ij} units of raw material j ($i = 1, \dots, k, j = 1, \dots, r$). Raw materials are bought in bundles that combine different types and quantities. Bundle h contains b_{hj} units of raw material j . There are s different bundles on the market. The bundles are unique, thus they can be bought at most once. The cost of bundle h is c_h euros. Each unit of feed i returns g_i euros. Formulate the problem of planning the production of feeds and the purchase of raw material bundles so as to maximize the profit (return - cost). The model must be linear. Explain the meaning of variables and constraints.

Variante: Let us assume that the company has to set up the inventory for the raw materials keeping them separate. The cost for setting up the inventory for raw material j is F_j , and it is independent from the quantity that is stored, but it is 0 if raw material j is not needed by the production. Thus if raw material j is needed, the company has to invest F_j euros for the inventory. How does the formulation change?

6. A family must do a very long trip on the highway without exiting, going from A to B. In the (unique) itinerary they pass by n gas stations. When they leave from A the gas tank is full at the maximum capacity (U liters) and this quantity is not sufficient to reach the destination. Therefore, they must stop during the trip at some gas stations to refuel. For the sake of simplicity, the refueling can be done in such a way that each time the upper capacity limit of U is reached. Let c_i be the gasoline cost per liter at station $i, i = 1, \dots, n$, moreover let p_i the gasoline consumption in number of liters to cover the distance from station $i - 1$ to station $i, i = 1, \dots, n + 1$. In order to simplify the notation let us assume that $A=0$ and $B=n + 1$.

Formulate the problem of determining where to refuel and the purchased quantity of gasoline as a shortest path problem on a suitable graph. Describe the nodes, the arcs and their costs. What is the main property of the graph? What is the most efficient algorithm to be used to solve the problem? Justify your answers.